Acceptance probabilities of rejection samplers for r-Pareto processes

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This note collects (a) the acceptance probability of the sampler that turns r_p-Pareto draws into r_0-Pareto draws, which underlies Theorem 1 of the manuscript, and (b) the acceptance probability of the convex-set sampler of Theorem 5, which turns r_0-Pareto draws into r-Pareto draws for a convex risk functional r. Both are instances of one rejection step (Lemma 2), and for L_p norms their acceptance probabilities multiply to $D^{1/p-1}$ (Corollary 1).

1 Setting

Let $\Omega = \mathbb{R}_+^D \setminus \{0\}$ and let κ be an exponent measure on Ω, homogeneous of order -1, i.e., $\kappa(tB) = t^{-1}\kappa(B)$ for $t > 0$, with unit Pareto margins, $\kappa(\{\boldsymbol{x} \in \Omega : x_j > 1\}) = 1$, $j = 1, \dots, D$. A risk functional is a measurable $r : \Omega \to [0, \infty)$, homogeneous of order 1, with $0 < \kappa(\mathcal{A}_r) < \infty$, where $\mathcal{A}_r = \{\boldsymbol{x} \in \Omega : r(\boldsymbol{x}) > 1\}$. The r-Pareto random vector $\tilde{\boldsymbol{Z}}^{(r)}$ has distribution $\kappa(\cdot \cap \mathcal{A}_r)/\kappa(\mathcal{A}_r)$, and all r-Pareto vectors below share the same κ. As in the manuscript, $r_0(\boldsymbol{x}) = D^{-1}\|\boldsymbol{x}\|_1$ is the spatial mean, $r_p(\boldsymbol{x}) = (D^{-1}\sum_{j=1}^D x_j^p)^{1/p} = D^{-1/p}\|\boldsymbol{x}\|_p$ for $p \in (1, \infty)$ and $r_\infty(\boldsymbol{x}) = \|\boldsymbol{x}\|_\infty$, with the convention $1/\infty = 0$. We write $\mathcal{A}_0 = \mathcal{A}_{r_0}$ and $\mathcal{A}_p = \mathcal{A}_{r_p}$.

Write $\boldsymbol{x} = s\boldsymbol{w}$ with $s = \|\boldsymbol{x}\|_1$ and $\boldsymbol{w} \in \Delta = \{\boldsymbol{w} \in \mathbb{R}_+^D : \|\boldsymbol{w}\|_1 = 1\}$, so that $\kappa(\mathrm{d}s, \mathrm{d}\boldsymbol{w}) = s^{-2}\mathrm{d}s\, H(\mathrm{d}\boldsymbol{w})$, where H is the angular measure of κ.

(F1) Since $x_j > 1$ if and only if $s > 1/w_j$, unit Pareto margins are equivalent to

$$1 = \int_\Delta \int_{1/w_j}^\infty s^{-2}\mathrm{d}s\, H(\mathrm{d}\boldsymbol{w}) = \int_\Delta w_j\, H(\mathrm{d}\boldsymbol{w}), \qquad j = 1, \dots, D.$$

Hence $H(\Delta) = \sum_{j=1}^D \int_\Delta w_j\, H(\mathrm{d}\boldsymbol{w}) = D$ and $\kappa(\mathcal{A}_0) = \int_\Delta \int_D^\infty s^{-2}\mathrm{d}s\, H(\mathrm{d}\boldsymbol{w}) = H(\Delta)/D = 1$.

(F2) For $t \geq 1$ and Borel $B \subseteq \Delta$, $\Pr\{r_0(\tilde{\boldsymbol{Z}}^{(r_0)}) > t,\ \tilde{\boldsymbol{Z}}^{(r_0)}/\|\tilde{\boldsymbol{Z}}^{(r_0)}\|_1 \in B\} = (Dt)^{-1}H(B) = t^{-1}H(B)/D$. Thus $\tilde{\boldsymbol{Z}}^{(r_0)} = DR\boldsymbol{W}$, where $R = r_0(\tilde{\boldsymbol{Z}}^{(r_0)})$ is unit Pareto, $\boldsymbol{W} = \tilde{\boldsymbol{Z}}^{(r_0)}/\|\tilde{\boldsymbol{Z}}^{(r_0)}\|_1 \sim H/D$, R and $\boldsymbol{W}$ are independent, and $\mathrm{E}\boldsymbol{W} = D^{-1}\mathbf{1}$ by (F1).

Lemma 1. *For any risk functional r, $\kappa(\mathcal{A}_r) = \int_\Delta r(\boldsymbol{w})\, H(d\boldsymbol{w}) = D\,\mathrm{E}\, r(\boldsymbol{W})$.*

Proof. Since $r(s\boldsymbol{w}) = s\, r(\boldsymbol{w})$, we have $s\boldsymbol{w} \in \mathcal{A}_r$ if and only if $s > 1/r(\boldsymbol{w})$ (no s if $r(\boldsymbol{w}) = 0$), so $\kappa(\mathcal{A}_r) = \int_\Delta \int_{1/r(\boldsymbol{w})}^\infty s^{-2}\mathrm{d}s\, H(\mathrm{d}\boldsymbol{w}) = \int_\Delta r(\boldsymbol{w})\, H(\mathrm{d}\boldsymbol{w})$. The second equality follows from (F2). □

2 A generic rejection step

Lemma 2. *Let r_a and r_b be risk functionals and $t > 0$ such that*

$$t\mathcal{A}_{r_b} = \{\boldsymbol{x} \in \Omega : r_b(\boldsymbol{x}) > t\} \subseteq \mathcal{A}_{r_a}. \tag{1}$$

Then

(a) $t^{-1}\tilde{\boldsymbol{Z}}^{(r_a)} \mid r_b\big(\tilde{\boldsymbol{Z}}^{(r_a)}\big) > t \overset{\mathrm{d}}{=} \tilde{\boldsymbol{Z}}^{(r_b)}$;

(b) $\Pr\big\{r_b\big(\tilde{\boldsymbol{Z}}^{(r_a)}\big) > t\big\} = t^{-1}\kappa(\mathcal{A}_{r_b})/\kappa(\mathcal{A}_{r_a})$.

Moreover, (1) holds if and only if $r_b \le t\, r_a$ on Ω, i.e., if and only if $t \ge M_{ba} := \sup\{r_b(\boldsymbol{x}) : \boldsymbol{x} \in \Omega,\ r_a(\boldsymbol{x}) \le 1\}$. Since (b) is decreasing in t, the largest acceptance probability is obtained with $t = M_{ba}$.

Proof. For Borel $B \subseteq \Omega$, $\{t^{-1}\tilde{\boldsymbol{Z}}^{(r_a)} \in B,\ r_b(\tilde{\boldsymbol{Z}}^{(r_a)}) > t\} = \{\tilde{\boldsymbol{Z}}^{(r_a)} \in t(B \cap \mathcal{A}_{r_b})\}$, and $t(B \cap \mathcal{A}_{r_b}) \subseteq t\mathcal{A}_{r_b} \subseteq \mathcal{A}_{r_a}$ by (1). By homogeneity of κ,

$$\Pr\big\{t^{-1}\tilde{\boldsymbol{Z}}^{(r_a)} \in B,\ r_b\big(\tilde{\boldsymbol{Z}}^{(r_a)}\big) > t\big\} = \frac{\kappa\big(t(B \cap \mathcal{A}_{r_b}) \cap \mathcal{A}_{r_a}\big)}{\kappa(\mathcal{A}_{r_a})} = \frac{\kappa\big(t(B \cap \mathcal{A}_{r_b})\big)}{\kappa(\mathcal{A}_{r_a})} = t^{-1}\frac{\kappa(B \cap \mathcal{A}_{r_b})}{\kappa(\mathcal{A}_{r_a})}.$$

Taking $B = \Omega$ gives (b), and dividing by (b) gives $\kappa(B \cap \mathcal{A}_{r_b})/\kappa(\mathcal{A}_{r_b}) = \Pr\{\tilde{\boldsymbol{Z}}^{(r_b)} \in B\}$, i.e., (a). For the equivalence, (1) states that $r_a(\boldsymbol{x}) \le 1$ implies $r_b(\boldsymbol{x}) \le t$. If $r_a(\boldsymbol{x}) > 0$, applying this to $\boldsymbol{x}/r_a(\boldsymbol{x})$ gives $r_b(\boldsymbol{x}) \le t\, r_a(\boldsymbol{x})$; if $r_a(\boldsymbol{x}) = 0$, applying it to $\lambda\boldsymbol{x}$, $\lambda \to \infty$, gives $r_b(\boldsymbol{x}) = 0$. The converse is immediate. □

Remark 1. *(i) If κ is homogeneous of order $-\alpha$, $\alpha > 0$, the same proof gives (a), and (b) with $t^{-\alpha}$ in place of t^{-1}. (ii) If $t < M_{ba}$, there exists $\boldsymbol{x}$ with $r_a(\boldsymbol{x}) \le 1$ and $r_b(\boldsymbol{x}) > t$, and the accepted draws miss $t\mathcal{A}_{r_b} \setminus \mathcal{A}_{r_a}$, which is charged by some κ with unit Pareto margins (e.g., one whose angular measure has an atom at $\boldsymbol{x}/\|\boldsymbol{x}\|_1$). Hence $t \ge M_{ba}$ is needed for the sampler to be exact for every such κ. (iii) Theorem 1 is Lemma 2 with $(r_a, r_b, t) = (r_p, r_0, u)$; the convex-set sampler of Theorem 5 has $(r_a, r_b, t) = (r_0, r, M)$ with $M = D \max_i r(\boldsymbol{e}_i)$; and Dombry et al. (2024, Proposition 2) has $(r_a, r_b, t) = (r_0, r, M')$ for any M' with $r \le M' r_0$.*

3 From r_p-Pareto to r_0-Pareto (Theorem 1)

Proposition 1. *Let $p \in (1, \infty]$ and $u \ge 1$.*

(i) $u^{-1}\tilde{\boldsymbol{Z}}^{(r_p)} \mid r_0\big(\tilde{\boldsymbol{Z}}^{(r_p)}\big) > u \overset{\mathrm{d}}{=} \tilde{\boldsymbol{Z}}^{(r_0)}$.

(ii) $\rho_p(u) := \Pr\big\{r_0\big(\tilde{\boldsymbol{Z}}^{(r_p)}\big) > u\big\} = u^{-1}\kappa(\mathcal{A}_0)/\kappa(\mathcal{A}_p) = u^{-1}/\kappa(\mathcal{A}_p)$, *which is decreasing in u. The choice $u = 1$ maximises it, giving*

$$\rho_p := \rho_p(1) = \frac{1}{\kappa(\mathcal{A}_p)} = \frac{D^{1/p-1}}{\mathrm{E}\|\boldsymbol{W}\|_p}.$$

(iii) $\kappa(\mathcal{A}_p) = \int_\Delta r_p(\boldsymbol{w})\, H(d\boldsymbol{w}) = D^{1-1/p}\, \mathrm{E}\|\boldsymbol{W}\|_p$ *and* $1 \le \kappa(\mathcal{A}_p) \le D^{1-1/p}$, *so that* $\rho_p \in [D^{1/p-1}, 1]$. *The value* $\rho_p = 1$ *is attained under complete dependence,* $H = D\,\delta_{D^{-1}\mathbf{1}}$, *and* $\rho_p = D^{1/p-1}$ *under asymptotic independence,* $H = \sum_{j=1}^{D} \delta_{\boldsymbol{e}_j}$.

Proof. By the power mean (or Jensen's) inequality, $r_0 \le r_p$, with equality at $\boldsymbol{x} = \mathbf{1}$, so $M_{ba} = 1$ in Lemma 2 with $(r_a, r_b) = (r_p, r_0)$. Explicitly, for $u \ge 1$, $r_0(\boldsymbol{x}) > u$ implies $r_p(\boldsymbol{x}) \ge r_0(\boldsymbol{x}) > u \ge 1$, i.e., $u\mathcal{A}_0 \subseteq \mathcal{A}_p$. Lemma 2 then gives (i) and the first expression in (ii), and $\kappa(\mathcal{A}_0) = 1$ by (F1). The last expression in (ii) follows from (iii). For (iii), Lemma 1 with $r = r_p$ gives the integral representation, and $r_p(\boldsymbol{w}) = D^{-1/p}\|\boldsymbol{w}\|_p$. For $\boldsymbol{w} \in \Delta$, $D^{-1} = r_0(\boldsymbol{w}) \le r_p(\boldsymbol{w}) \le D^{-1/p}\|\boldsymbol{w}\|_1 = D^{-1/p}$, and integrating against H, with $H(\Delta) = D$, gives $1 \le \kappa(\mathcal{A}_p) \le D^{1-1/p}$. Equality holds in the lower bound when H is concentrated on $D^{-1}\mathbf{1}$, where $r_p(D^{-1}\mathbf{1}) = D^{-1}$, and in the upper bound when H is concentrated on the vertices, where $r_p(\boldsymbol{e}_j) = D^{-1/p}$. □

Remark 2. *For* $p = \infty$, $\kappa(\mathcal{A}_\infty) = \kappa\big([\mathbf{0}, \mathbf{1}]^c\big) = V(\mathbf{1}) = \theta_D$, *the* D*-variate extremal coefficient. Hence* $\rho_\infty = 1/\theta_D$*: the proportion of exceedances of the maximum that are retained for the spectral likelihood is the reciprocal of the* D*-variate extremal coefficient.*

4 From r_0-Pareto to r-Pareto with convex r (Theorem 5)

As in Theorem 5, let r be convex and homogeneous of order 1, with $r(\mathbf{0}) = 0$, and let $M = D \max_i r(\boldsymbol{e}_i) = \max_i r(D\boldsymbol{e}_i) > 0$.

Lemma 3. *r is subadditive, and for all $\boldsymbol{w} \in \Delta$,*

$$r(\boldsymbol{w}) \le \sum_{j=1}^{D} w_j\, r(\boldsymbol{e}_j) \le \max_j r(\boldsymbol{e}_j) = \frac{M}{D}. \tag{2}$$

Consequently, $r \le M r_0$ on Ω, so (1) holds with $(r_a, r_b, t) = (r_0, r, M)$. Since the bound (2) is attained at a vertex, $M_{ba} = M$.

Proof. Convexity and homogeneity give $r(\boldsymbol{x} + \boldsymbol{y}) = 2\, r\{(\boldsymbol{x} + \boldsymbol{y})/2\} \le r(\boldsymbol{x}) + r(\boldsymbol{y})$. Hence, using $r(\mathbf{0}) = 0$ when $w_j = 0$, $r(\boldsymbol{w}) = r\big(\sum_j w_j \boldsymbol{e}_j\big) \le \sum_j r(w_j \boldsymbol{e}_j) = \sum_j w_j\, r(\boldsymbol{e}_j)$. The second inequality in (2) holds since $\boldsymbol{w} \in \Delta$, and $r(\boldsymbol{x}) = \|\boldsymbol{x}\|_1\, r(\boldsymbol{x}/\|\boldsymbol{x}\|_1) \le \|\boldsymbol{x}\|_1 M/D = M r_0(\boldsymbol{x})$ for $\boldsymbol{x} \in \Omega$. □

Lemma 3 is the analytic form of the convex-set argument in Section E.1 of the Supplemental Material, namely $\mathcal{A}^c_{r_0} = \mathrm{conv}\{\mathbf{0}, D\boldsymbol{e}_1, \ldots, D\boldsymbol{e}_D\} \subseteq \mathcal{A}^c_{r/M}$.

Proposition 2. *Under the above conditions, with* $\boldsymbol{W} = \tilde{\boldsymbol{Z}}^{(r_0)}/\|\tilde{\boldsymbol{Z}}^{(r_0)}\|_1 \sim H/D$*:*

(i) $\tilde{\boldsymbol{Z}}^{(r_0)}/M \mid r\big(\tilde{\boldsymbol{Z}}^{(r_0)}\big) > M \overset{\mathrm{d}}{=} \tilde{\boldsymbol{Z}}^{(r)}$;

(ii) the acceptance probability is

$$\pi_r := \Pr\big\{r\big(\tilde{\boldsymbol{Z}}^{(r_0)}\big) > M\big\} = \frac{\kappa(\mathcal{A}_r)}{M\,\kappa(\mathcal{A}_0)} = \frac{1}{M}\int_\Delta r(\boldsymbol{w})\, H(d\boldsymbol{w}) = \frac{\mathrm{E}\, r(\boldsymbol{W})}{\max_j r(\boldsymbol{e}_j)};$$

(iii) $\dfrac{r(\mathbf{1})}{D\max_j r(\boldsymbol{e}_j)} \le \pi_r \le \dfrac{\sum_{j=1}^D r(\boldsymbol{e}_j)}{D\max_j r(\boldsymbol{e}_j)} \le 1$. *The lower bound is attained under complete dependence and the upper bound under asymptotic independence, so both bounds are sharp over exponent measures with unit Pareto margins;*

(iv) for $r = r_p$, $p \in (1,\infty]$, $M = D^{1-1/p}$ *and*

$$\pi_p := \Pr\{r_p(\tilde{\boldsymbol{Z}}^{(r_0)}) > D^{1-1/p}\} = \frac{\kappa(\mathcal{A}_p)}{D^{1-1/p}} = \mathrm{E}\|\boldsymbol{W}\|_p \in [D^{1/p-1}, 1],$$

with $\pi_p = D^{1/p-1}$ *under complete dependence and* $\pi_p = 1$ *under asymptotic independence; for* $p = \infty$, $\pi_\infty = \theta_D/D$*;*

(v) for $r(\boldsymbol{x}) = \sum_j m_j x_j$, $m_j \ge 0$, $\max_j m_j > 0$, $M = D\max_j m_j$ *and* $\pi_r = \overline{m}/\max_j m_j$ *with* $\overline{m} = D^{-1}\sum_j m_j$*, whatever the dependence structure.*

Proof. (i)–(ii) By Lemma 3, (1) holds with $(r_a, r_b, t) = (r_0, r, M)$, and Lemma 2 gives (i) and the first expression in (ii). The others follow from $\kappa(\mathcal{A}_0) = 1$ (F1) and Lemma 1. Alternatively, by (F2), $\Pr\{DR\, r(\boldsymbol{W}) > M \mid \boldsymbol{W}\} = \min\{1, D\, r(\boldsymbol{W})/M\} = D\, r(\boldsymbol{W})/M$, since $D\, r(\boldsymbol{W}) \le M$ by (2).

(iii) Since r is convex and $\mathrm{E}\boldsymbol{W} = D^{-1}\mathbf{1}$ (F2), Jensen's inequality gives $\mathrm{E}\, r(\boldsymbol{W}) \ge r(D^{-1}\mathbf{1}) = r(\mathbf{1})/D$. Taking expectations in (2) gives $\mathrm{E}\, r(\boldsymbol{W}) \le \sum_j r(\boldsymbol{e}_j)\mathrm{E}W_j = D^{-1}\sum_j r(\boldsymbol{e}_j) \le \max_j r(\boldsymbol{e}_j)$. Under complete dependence, $\boldsymbol{W} = D^{-1}\mathbf{1}$ almost surely and Jensen's inequality is an equality; under asymptotic independence, $\boldsymbol{W}$ is uniform on $\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_D\}$ and the first inequality in (2) is an equality almost surely.

(iv) $r_p(\boldsymbol{e}_j) = D^{-1/p}$ gives $M = D^{1-1/p}$, $r_p(\mathbf{1}) = 1$, and $\kappa(\mathcal{A}_p) = D^{1-1/p}\mathrm{E}\|\boldsymbol{W}\|_p$ by Proposition 1(iii); the bounds follow from (iii), and the case $p = \infty$ from Remark 2.

(v) $r(\boldsymbol{W})$ is linear in $\boldsymbol{W}$, so $\mathrm{E}\, r(\boldsymbol{W}) = \sum_j m_j \mathrm{E}W_j = \overline{m}$. □

Remark 3 (Comparison with Dombry *et al.*, 2024, and Table 1)**.** *By Lemma 2 with* $(r_a, r_b, t) = (r_0, r, M')$*, the sampler of Dombry* et al. *(2024, Proposition 2) has acceptance probability* $D\,\mathrm{E}\, r(\boldsymbol{W})/M'$*, so its ratio to* π_r *is* M/M' *whatever the dependence structure. For* $r = r_p$ *with* $M' = D^{p-1-1/p}$*, which is the constant* D^{p-2} *of Dombry* et al. *(2024) for* $\|\cdot\|_p \le D^{p-2}\|\cdot\|_1$ *rescaled to* r_p *and* r_0*, the ratio is* $D^{1-1/p}/D^{p-1-1/p} = D^{2-p}$*, which matches Table 1: for* $p = 3$, $52/4 = 13.0$, $29/16 = 1.81$ *and* $18/64 = 0.28$ *(tabulated* 13, 1.80, 0.27*); for* $p = 5$, $D = 4$, $49/64 = 0.77$ *(tabulated* 0.76*); and for* $p = 2$ *the two samplers coincide. For* $(p, D) = (5, 16)$ *and* $(10, 4)$ *the ratio gives* $26/16^3 \approx 0.006$ *and* $47/4^8 \approx 0.0007$*, so these entries are below* 0.01%*.*

5 Complementarity of the two samplers

Corollary 1. *For* $p \in (1,\infty]$ *and every exponent measure with unit Pareto margins,*

$$\rho_p\, \pi_p = D^{1/p-1},$$

where ρ_p *is the acceptance probability of the* $r_p \to r_0$ *sampler with* $u = 1$ *(Proposition 1) and* π_p *is that of the* $r_0 \to r_p$ *sampler (Proposition 2(iv)). Complete dependence gives* $(\rho_p, \pi_p) = (1, D^{1/p-1})$*, and asymptotic independence gives* $(\rho_p, \pi_p) = (D^{1/p-1}, 1)$*.*

Proof. $\rho_p = 1/\kappa(\mathcal{A}_p)$ and $\pi_p = \kappa(\mathcal{A}_p)/D^{1-1/p}$. □

Remark 4. *(i) The empirical acceptance rate of the simulation step is a Monte Carlo estimate of* $\pi_p = \mathrm{E}\|\boldsymbol{W}\|_p$*, so the proportion of* r_p *exceedances retained by the spectral likelihood under the simulated model follows at no extra cost as* $\rho_p = D^{1/p-1}/\pi_p$*. For the Brown–Resnick model of Table 1 with* $D = 100$*,* $\pi_3 \approx 0.16$ *gives* $\rho_3 \approx 100^{-2/3}/0.16 \approx 0.29$*, and* $\pi_{10} \approx 0.13$ *gives* $\rho_{10} \approx 100^{-0.9}/0.13 \approx 0.12$*.*

(ii) More generally, for a convex r *with* $m_r := \inf_{\boldsymbol{w}\in\Delta} r(\boldsymbol{w}) > 0$*, the sharpest constant in* $r_0 \le u\, r$ *is* $M_{ba} = 1/(Dm_r)$ *(Lemma 2 with* $(r_a, r_b) = (r, r_0)$*). The* $r \to r_0$ *sampler with* $u = 1/(Dm_r)$ *then has acceptance probability* $\rho_r = m_r/\mathrm{E}\, r(\boldsymbol{W})$ *by Lemma 2(b) and Lemma 1, and* $\rho_r\pi_r = m_r/\max_j r(\boldsymbol{e}_j)$*. For* $r = r_p$*,* $m_r = D^{-1}$ *and* $\max_j r(\boldsymbol{e}_j) = D^{-1/p}$*, which recovers Corollary 1.*

References

Dombry, C., Legrand, J. and Opitz, T. (2024) Pareto processes for threshold exceedances in spatial extremes. *arXiv preprint arXiv:2407.05699* .